8 **(a)** The dense core emits white light [1]
As the white light passes through the cooler gas layer, photons of energy corresponding to the energy level differences in the gas are absorbed [1]

(b) Since the wavelength corresponding to Hydrogen are darker/thicker [1]
There is more hydrogen than helium [1]

(c) Since there are more lines present in Fig 8.1 than there are present in Fig 8.2 and 8.3, [M1 – comparing the spectra]

These lines are likely caused by other gases. [A1 – other lines indicate other gases]

(d) **(i)** $c = f\lambda$

$$f = \frac{3\times10^{8}}{410\times10^{-9}} \quad [\text{B1}]$$
$$= 7.3\times10^{14}\ \text{Hz}$$

(ii) $E = hf$

$$= (6.63\times10^{-34})(7.3\times10^{14}) \quad [\text{M1}]$$
$$= 4.84\times10^{-19}\ \text{J} \quad [\text{A1}]$$

(e) **(i)**

Hydrogen spectral wavelengths from GN-Z11 / nm	Hydrogen spectral wavelengths from stationary source on Earth / nm	z	v / m s^{-1}
676	656	0.0305	9.15×10^{6}
500	485	0.0309	9.27×10^{6}

$$z = \frac{500-485}{485} = 0.0309$$

$$v = 0.0309\times(3.0\times10^{8})$$
$$= 9.27\times10^{6}\ \text{m s}^{-1}$$

[B1 per cell]

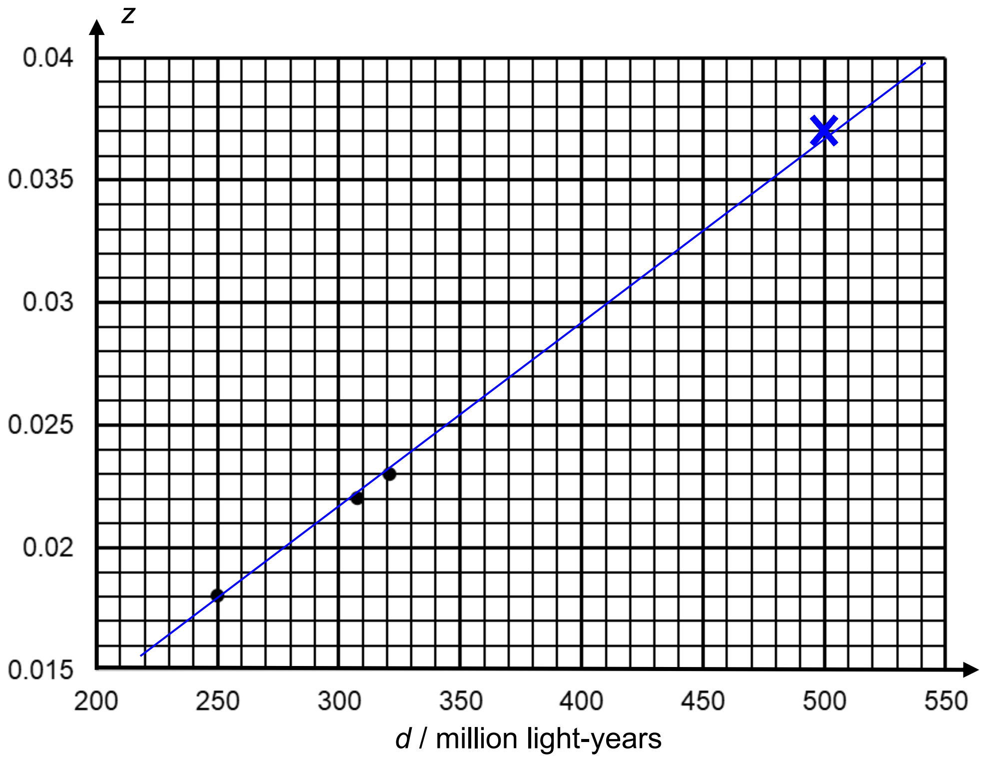
(ii) Since the wavelengths become longer,
They are closer to the red end of the visible spectrum [B1]
Hence they are called “red-shift”

(iii) The Andromeda Galaxy is moving toward Earth.

(f) **(i)** $\text{one lightyear} = 365\times24\times60\times60\times(3.0\times10^{8})$
$$= 9.46\times10^{15}\ \text{m}$$

(ii) The further the galaxy, the faster it is moving away from the Earth.

(iii)



Plot [B1]
Best fit line [B1]

(iv) Using (0.039, 530) and (0.018, 250)

$$\frac{H_o}{c} = \frac{(0.039 - 0.018)}{(530 - 250) \times (9.46 \times 10^{15} \times 10^{6})}$$

$$= 7.93 \times 10^{-27} \quad [C1]$$

$$H_o = (3 \times 10^{8})(7.93 \times 10^{-27}) \quad [C1]$$

$$= 2.38 \times 10^{-18} \quad [A1]$$